

Dot Product

Exercise 1

$$\begin{array}{cc} & a \\ & \boxed{5} \\ b & \boxed{3} \quad \boxed{} \\ & a \cdot b \end{array}$$

Dot Product

Exercise 2

$$\begin{array}{cc} & a \\ & \boxed{2} \\ b & \boxed{4} \quad \boxed{} \\ & a \cdot b \end{array}$$

Dot Product

Exercise 3

$$\begin{array}{cc} & a \\ & \boxed{2} \\ b & \boxed{-3} \end{array} \quad \begin{array}{c} \\ \\ \boxed{} \\ a \cdot b \end{array}$$

Dot Product

Exercise 4

$$\begin{array}{cc} & a \\ & \boxed{2} \\ b & \boxed{} \\ & \boxed{6} \\ & a \cdot b \end{array}$$

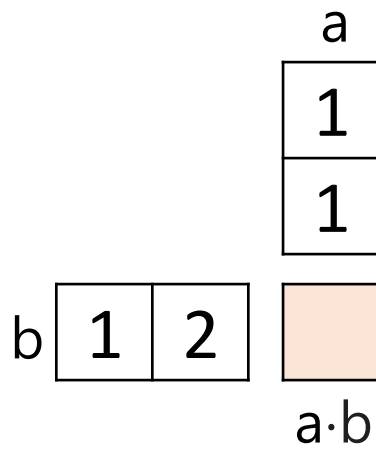
Dot Product

Exercise 5

$$\begin{array}{cc} & a \\ & \square \\ b & \begin{array}{c} \square \\ -3 \end{array} & \begin{array}{c} \square \\ 9 \end{array} \\ & a \cdot b \end{array}$$

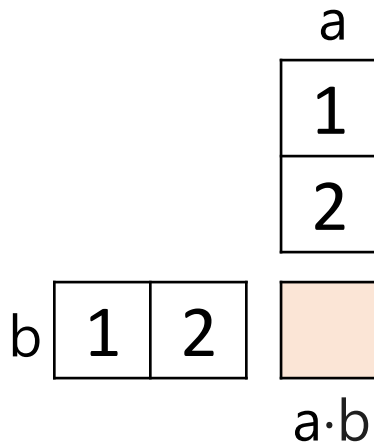
Dot Product

Exercise 6



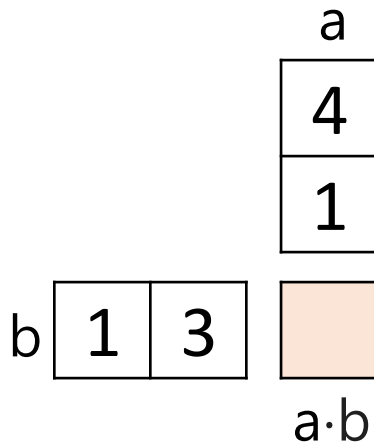
Dot Product

Exercise 7



Dot Product

Exercise 8



Dot Product

Exercise 9

$$\begin{array}{c} a \\ \begin{array}{|c|} \hline 4 \\ \hline \end{array} \\ \begin{array}{|c|} \hline 1 \\ \hline \end{array} \\ \begin{array}{|c|} \hline \text{ } \\ \hline \end{array} \\ a \cdot b \end{array} \quad \begin{array}{c} b \\ \begin{array}{|c|c|} \hline 1 & -1 \\ \hline \end{array} \end{array}$$

Dot Product

Exercise 10

$$\begin{array}{c} a \\ \begin{array}{|c|} \hline 4 \\ \hline \end{array} \\ \begin{array}{|c|} \hline 1 \\ \hline \end{array} \\ \begin{array}{|c|} \hline \text{ } \\ \hline \end{array} \\ a \cdot b \end{array} \quad \begin{array}{c} b \\ \begin{array}{|c|c|} \hline 0 & -1 \\ \hline \end{array} \end{array}$$

Dot Product

Exercise 11

$$\begin{array}{c} a \\ \begin{array}{|c|} \hline -4 \\ \hline \end{array} \\ \begin{array}{|c|} \hline 2 \\ \hline \end{array} \\ \begin{array}{|c|} \hline \text{ } \\ \hline \end{array} \\ a \cdot b \end{array} \quad \begin{array}{c} b \\ \begin{array}{|c|c|} \hline 0 & -3 \\ \hline \end{array} \end{array}$$

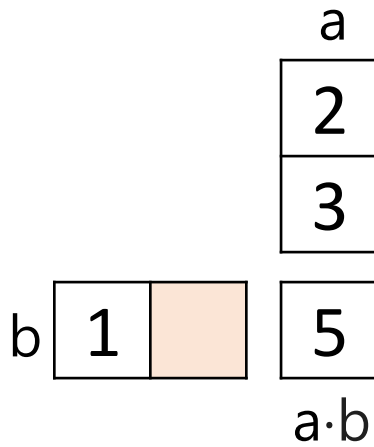
Dot Product

Exercise 12

$$\begin{array}{c} a \\ \begin{array}{|c|} \hline -4 \\ \hline 0 \\ \hline \end{array} \\ b \begin{array}{|c|c|} \hline 0 & -3 \\ \hline \end{array} \quad \begin{array}{|c|} \hline \text{ } \\ \hline \end{array} \\ a \cdot b \end{array}$$

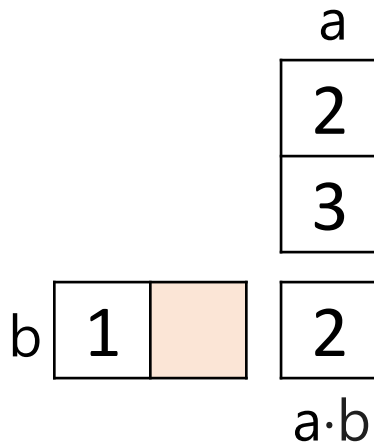
Dot Product

Exercise 13



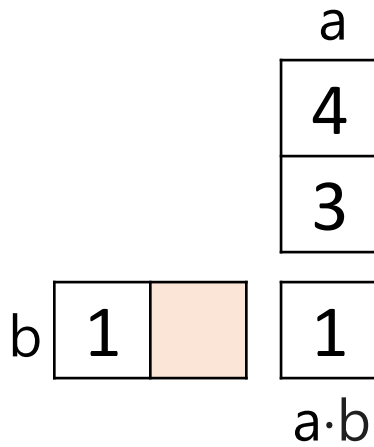
Dot Product

Exercise 14



Dot Product

Exercise 15



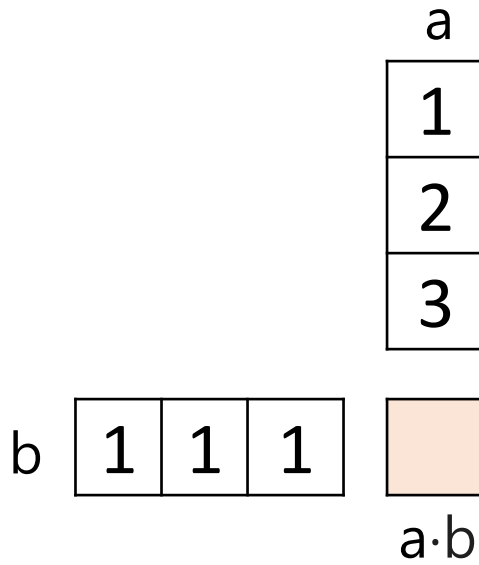
Dot Product

Exercise 16

$$\begin{array}{cc} & a \\ & \begin{array}{|c|} \hline 2 \\ \hline \end{array} \\ & \begin{array}{|c|} \hline 3 \\ \hline \end{array} \\ b & \begin{array}{|c|c|} \hline 1 & \\ \hline \end{array} & \begin{array}{|c|} \hline 8 \\ \hline \end{array} \\ & & a \cdot b \end{array}$$

Dot Product

Exercise 17



Dot Product

Exercise 18

$$\begin{array}{c} a \\ \begin{array}{|c|} \hline 4 \\ \hline 9 \\ \hline 3 \\ \hline \end{array} \\ \\ b \quad \begin{array}{|c|c|c|} \hline 1 & 0 & -1 \\ \hline \end{array} \quad \begin{array}{|c|} \hline \text{ } \\ \hline \end{array} \\ a \cdot b \end{array}$$

Dot Product

Exercise 19

$$\begin{array}{c} \mathbf{a} \\ \begin{array}{|c|} \hline 0 \\ \hline 1 \\ \hline 0 \\ \hline \end{array} \end{array}$$
$$\begin{array}{c} \mathbf{b} \\ \begin{array}{|c|c|c|} \hline 1 & 0 & -1 \\ \hline \end{array} \end{array}$$
$$\begin{array}{c} \text{a} \cdot \text{b} \\ \begin{array}{|c|} \hline \text{ } \\ \hline \end{array} \end{array}$$

Dot Product

Exercise 20

$$\begin{array}{c} \mathbf{a} \\ \begin{array}{|c|} \hline 8 \\ \hline 4 \\ \hline 5 \\ \hline \end{array} \end{array} \quad \begin{array}{c} \mathbf{b} \\ \begin{array}{|c|c|c|} \hline 1 & 0 & \\ \hline \end{array} \end{array} \quad \begin{array}{c} \phantom{\mathbf{a}} \\ \begin{array}{|c|} \hline 3 \\ \hline \end{array} \end{array}$$

$\mathbf{a} \cdot \mathbf{b}$

Dot Product

Exercise 21

$$\begin{array}{c} \mathbf{a} \\ \begin{array}{|c|} \hline 4 \\ \hline 2 \\ \hline 9 \\ \hline \end{array} \end{array} \quad \begin{array}{c} \mathbf{b} \\ \begin{array}{|c|c|c|} \hline \text{ } & 1 & 0 \\ \hline \end{array} \end{array} \quad \begin{array}{|c|} \hline 6 \\ \hline \end{array}$$

$\mathbf{a \cdot b}$

Dot Product

Exercise 22

$$\begin{array}{c} \mathbf{a} \\ \begin{array}{|c|} \hline 4 \\ \hline 2 \\ \hline 9 \\ \hline \end{array} \end{array}$$
$$\begin{array}{c} \mathbf{b} \\ \begin{array}{|c|c|c|} \hline \text{ } & 1 & 0 \\ \hline \end{array} \end{array}$$
$$\begin{array}{c} -2 \\ \hline \end{array}$$
$$\mathbf{a} \cdot \mathbf{b}$$

Dot Product

Exercise 23

$$\begin{array}{c} \mathbf{a} \\ \begin{array}{|c|} \hline 1 \\ \hline 1 \\ \hline \end{array} \\ \\ \mathbf{b} \quad \begin{array}{|c|c|c|} \hline 3 & 1 & -2 \\ \hline \end{array} \quad \begin{array}{|c|} \hline 6 \\ \hline \end{array} \\ \mathbf{a \cdot b} \end{array}$$

Dot Product

Exercise 24

$$\begin{array}{c} \mathbf{a} \\ \begin{bmatrix} 1 \\ 4 \\ \end{bmatrix} \\ \mathbf{b} \quad \begin{bmatrix} 3 & 1 & -2 \end{bmatrix} \end{array} \quad \begin{array}{c} \\ \\ 7 \\ \mathbf{a \cdot b} \end{array}$$

Dot Product

Exercise 25

$$\begin{array}{c} \mathbf{a} \\ \begin{array}{|c|} \hline 1 \\ \hline 3 \\ \hline 0 \\ \hline \end{array} \end{array} \quad \begin{array}{c} \mathbf{b} \\ \begin{array}{|c|c|c|} \hline 2 & & -9 \\ \hline \end{array} \end{array} \quad \begin{array}{|c|} \hline -1 \\ \hline \end{array}$$

$\mathbf{a \cdot b}$